

Project Initialization and Planning Phase — AI WeatherWise

Date	17 September 2026
Team ID	SWTID-2026-2825
Project Name	AI WeatherWise
Maximum Marks	5 Marks

The proposal report aims to give users a secure, AI-assisted way to track weather for the places that matter to them — replacing raw metrics with personalized, natural-language guidance.

Project Overview

Objective	To provide users with a secure, centralized platform to track favorite weather locations and receive AI-generated, natural-language weather summaries and activity/clothing recommendations rather than raw meteorological data.
Scope	Deliver a RESTful backend (Node.js / Express / MongoDB) covering user authentication, favorite-location management, live weather retrieval via OpenWeatherMap, and AI-driven insight generation via Google Gemini, with resilient fallback handling.

Problem Statement

Description	Travellers and everyday users must manually check multiple weather sources and interpret raw metrics themselves, with no secure, personalized system to store preferences or translate data into actionable guidance.
Impact	Reduces the time and cognitive effort needed to plan around weather, improves user preparedness, and demonstrates a scalable pattern for combining live data APIs with generative AI.

Proposed Solution

Approach	Build a modular MVC backend where Controllers orchestrate OpenWeatherMap and Gemini calls while Mongoose Models manage User and Location persistence, secured end-to-end with JWT and bcrypt.
Key Features	- Secure JWT authentication & bcrypt password hashing - Favorite-location CRUD tied to the authenticated user - Real-time weather retrieval via OpenWeatherMap - Gemini-powered natural-language summaries & personalized recommendations - Resilient fallback mode when external API keys are unavailable - Centralized error handling & request sanitization

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU specification (no GPU required – this is a backend REST service, not a model-training workload)	Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or equivalent
Memory	RAM specification	8 GB minimum (16 GB recommended for running MongoDB, the Node server, and API clients concurrently)
Storage	Disk space for code, node_modules, and logs	1 GB available disk space
Software		
Frameworks	Node.js runtime & web framework	Node.js (v16+), Express.js
Libraries	Core dependencies	mongoose, bcryptjs, jsonwebtoken, cors, dotenv, multer, @google/genai
Development Environment	IDE / API client	Visual Studio Code, Thunder Client / Postman, nodemon (dev)
Data		
Data	Source, size, format	Live third-party APIs – OpenWeatherMap (current weather by city) & Google Gemini AI (generative text); persisted in MongoDB Users & Locations collections